

INDEPENDENT PEER-REVIEW PACKET

EC-010

Declared Convex-Combination Polynomial Identities

Cold Review Packet 5 of 20

Dylan M. Roy Mathematical Research Program

Prepared for independent validation - 27 July 2026

This packet is self-contained for mathematical review. Drive access is supplemental, not required to understand the claim and proof.

READ FIRST

You are being asked to review one narrow terminal object, not the wider q0/P0.2/Theorem B research program. The declared terminal label is an internal research-state classification. Your task is to decide whether the exact statement and its proof/evidence are correct within the stated hypotheses and exclusions.

1. Pass A - Independent reconstruction. Read the exact statement, notation, and hypotheses. Attempt your own derivation or falsification before reading the embedded historical review record.
2. Pass B - Source audit. Compare your reconstruction with the complete controlling source text embedded in Part II. Check every dependency, sign, factor, quantifier, scope boundary, and computational claim.
3. Pass C - Hostile review. Run the object-specific focal checks and invent at least one additional mutation or counterexample attempt.
4. Issue one fail-closed verdict: APPROVE; APPROVE WITH CLARIFICATIONS; AMEND REQUIRED; REJECT; or CANNOT VERIFY.

Do not transfer approval to excluded downstream claims. Do not count access to this packet, the author, or another reviewer as independent mathematical evidence.

Part I - Exact Review Object

Packet	5 of 20
Object ID	EC-010
Object class	Exact polynomial identities
Exact statement	The two displayed generic-chart and transverse-chart polynomial equalities hold identically; their convex-combination and automatic-nonnegativity interpretation is restricted to $0 \leq \alpha \leq 1$.
Internal terminal label	CLOSED - EXACT DECLARED CONVEX-COMBINATION IDENTITIES
Terminal date	2026-07-21
Recommended expertise	Algebra / polynomial identities / real analysis

1. Significance

This object was separated from larger theorem claims so that its correctness can be assessed without accepting unrelated Gaussian-field, persistence, adjacency, or asymptotic assertions. A positive review supports only the exact statement above and the dependency uses explicitly listed below.

2. Evidence class and prior review

Evidence	Coefficient-by-coefficient expansion and independent exact-rational reconstruction.
Prior review record	The equalities were independently expanded; broader fold implications were excluded.
Binding exclusions	No theorem about positivity outside α in $[0,1]$, fold nondegeneracy, or regional counting.

3. Dependency map and use restrictions

- Inputs: only definitions, hypotheses, and subordinate identities explicitly reproduced in the controlling source text.
- Output: only the exact statement listed in the review-object table.
- Permitted downstream use: as a cited lemma/interface under identical object definitions and hypotheses.
- Forbidden transfer: no approval of a parent theorem, numerical constant, external release, or canonical promotion unless separately reviewed.
- Object-specific exclusions: No theorem about positivity outside α in $[0,1]$, fold nondegeneracy, or regional counting.

4. Object-specific hostile checks

- Expand both sides in a computer algebra system or by hand.
- Separate equality from nonnegativity.
- Test alpha outside $[0,1]$ to confirm the scope firewall is necessary.
- Attempt at least one additional mutation that changes an orientation, sign, normalization, law, endpoint, or quantifier while leaving the text superficially similar.

5. Proof-obligation ledger

Obligation	Reviewer test	Initial state
O1 Object identity	Definitions, ordering, normalization, and law match the statement.	OPEN FOR REVIEW
O2 Hypotheses	Every hypothesis used is stated and no stronger premise is hidden.	OPEN FOR REVIEW
O3 Derivation	Each algebraic, analytic, topological, or probabilistic step is valid.	OPEN FOR REVIEW
O4 Quantifiers	Uniformity, existence, endpoint, and asymptotic quantifiers are correct.	OPEN FOR REVIEW
O5 Computation	Any executable or interval result is reproducible and non-vacuous.	OPEN FOR REVIEW
O6 Scope firewall	The conclusion is not silently enlarged beyond the exact object.	OPEN FOR REVIEW
O7 Contradictions	No known correction, counterexample, or retained caveat defeats the claim.	OPEN FOR REVIEW

Part II - Complete Controlling Source Text

The following is a paragraph-order-preserving text reproduction of the controlling Drive artifact. Typography is normalized for this packet; mathematical content is not intentionally rewritten. Use the manifest URL for native-file or byte-level checks.

GP-CLS-PROP-013-v1.0 — PROPOSED TERMINAL CLOSURE OF THE DECLARED CONVEX-COMBINATION IDENTITIES

Artifact ID: GP-CLS-PROP-013-v1.0

Author: OpenAI GPT-5.6 Thinking

Created: 2026-07-21

Class: CLS-PROP — additive terminal-closure proposal

Status: READY FOR HUMAN APPROVAL

Authority: none until Dylan M. Roy approves or modifies the terminal statement

Canonical impact: NONE before approval

1. Exact object card

OBJECT CLASS: DETERMINISTIC ALGEBRAIC IDENTITIES

SOURCE CONTEXT: generic and transverse local-chart brackets in the maximum–saddle fold analysis

VARIABLES: real α , K , c , η for Identity A; real α , a , η for Identity B

EVIDENCE CLASS: exact polynomial expansion

PROPOSED TERMINAL LABEL: CLOSED — EXACT DECLARED CONVEX-COMBINATION IDENTITIES

2. Exact terminal statements

Identity A — generic chart:

$$\begin{aligned} &(K + 8\alpha c\eta)^2 + 64\alpha(1-\alpha)c^2\eta^2 \\ &= (1-\alpha)K^2 + \alpha(K+8c\eta)^2. \end{aligned}$$

Identity B — transverse chart:

$$\begin{aligned} &(a + (1-2\alpha)\eta)^2 + 4\alpha(1-\alpha)\eta^2 \\ &= (1-\alpha)(a+\eta)^2 + \alpha(a-\eta)^2. \end{aligned}$$

Both identities hold for all real values of their declared variables. Their interpretation as convex combinations of endpoint squares, and the resulting automatic nonnegativity, applies when $0 \leq \alpha \leq 1$.

3. Fresh independent expansion

For Identity A, the left side expands as

$$\begin{aligned} &K^2 + 16\alpha Kc\eta + 64\alpha^2 c^2 \eta^2 + 64\alpha c^2 \eta^2 - 64\alpha^2 c^2 \eta^2 \\ &= K^2 + 16\alpha Kc\eta + 64\alpha c^2 \eta^2. \end{aligned}$$

The right side expands as

$$\begin{aligned} K^2 - \alpha K^2 + \alpha(K^2 + 16Kc\eta + 64c^2\eta^2) \\ = K^2 + 16\alpha Kc\eta + 64\alpha c^2\eta^2. \end{aligned}$$

For Identity B, the left side expands as

$$\begin{aligned} a^2 + 2a(1-2\alpha)\eta + (1-2\alpha)^2\eta^2 + 4\alpha(1-\alpha)\eta^2 \\ = a^2 + 2a\eta - 4\alpha a\eta + \eta^2. \end{aligned}$$

The right side expands as

$$\begin{aligned} (1-\alpha)(a^2+2a\eta+\eta^2) + \alpha(a^2-2a\eta+\eta^2) \\ = a^2 + 2a\eta - 4\alpha a\eta + \eta^2. \end{aligned}$$

Thus both identities are exact coefficient-by-coefficient polynomial equalities.

4. Evidence and review classification

Primary source: AI edit 0003 — 01_CORE_PAIRING_THEOREM.md, Review Note 1, Drive ID 1xMAeJYcjnqyFg16-sxnqjJi6MbB3cM-e.

Closure census: AO48-REG-018-v1.0 identified the independent expansion as the sole remaining step for S4.

Second-line route completed here: OpenAI expanded both sides independently from the displayed formulas and matched all coefficients. This is a separate exact algebraic re-execution, not an independent theorem proof and not evidence for the broader geometric interpretation.

5. Scope boundary and forbidden extrapolations

This proposed closure establishes only the two declared algebraic identities.

It does not by itself establish:

- • a third-critical-point no-go theorem;
- • strict positivity or strict negativity without the source-specific nondegeneracy assumptions;
- • the interpretation of α as a persistence-window coordinate in every chart;
- • quartic boundary-layer estimates;
- • integrable determinant-weighted bounds;
- • typed or adjacency-conditioned Palm estimates;
- • P0.2, Theorem B, a numerical constant, or any machine-root transition.

The phrase “convex combination” is restricted to $0 \leq \alpha \leq 1$. Outside that interval the equality remains exact, but the coefficients need not be nonnegative.

6. Dependencies affected

If approved, this closure:

- • retires repeated expansion of these two brackets;
- • permits exact citation of their endpoint-square interpolation form;
- • supports later local sign arguments only when all source conventions and additional hypotheses match;

- prevents the algebraic equality from being repeatedly treated as an unresolved proof obligation.

7. Dependencies explicitly unaffected

Unaffected:

- the validity of the surrounding fold theorem;
- the maximum/saddle classification and nondegeneracy conditions;
- finite-r remainder control;
- regional counting and defect dichotomy;
- typed-to-adjacent transfer;
- P0.2, Theorem B, canonical theorem status, and active q0 machine state.

8. Coupled-advancement no-change certificate

Mathematical status change proposed: two already-stated polynomial identities move from single-source review-note status to exact second-line-verified narrow closure.

Architectural change required: none beyond retaining α -domain and scope fields in the closure record.

Existing controls are sufficient because OP-PROT-004 requires exact object, evidence, affected and unaffected dependencies, human approval, and additive reopening. The current object-card discipline distinguishes exact algebra from the broader probabilistic and geometric claims.

Failure that would falsify this no-change judgment:

- a later artifact cites automatic nonnegativity outside $0 \leq \alpha \leq 1$;
- a different coefficient convention is silently substituted;
- the closed equality is cited as proof of a broader no-go or rate theorem.

9. Human approval requirement

Human approval is REQUIRED because terminal mathematical closure remains reserved to Dylan M. Roy.

Suggested approval statement:

“I approve GP-CLS-PROP-013-v1.0 exactly as a closure of the two declared polynomial identities and their convex-combination interpretation on $0 \leq \alpha \leq 1$. Record the terminal label CLOSED — EXACT DECLARED CONVEX-COMBINATION IDENTITIES, preserve all provenance, and do not infer any broader fold, Palm, rate, P0.2, Theorem B, or machine-state result from this closure.”

10. Additive correction and reopening path

Reopening requires new contradictory evidence identifying an algebraic error, a mismatched coefficient convention, or an invalid scope use. A correction must preserve this proposal and every source artifact, identify affected and unaffected dependencies, and reopen only this narrow identity unless broader contamination is demonstrated.

Part III - Terminal-State Reconciliation

1. Closure decision

Exact internal disposition	CLOSED - EXACT DECLARED CONVEX-COMBINATION IDENTITIES
Closure date	2026-07-21
Evidence basis	Coefficient-by-coefficient expansion and independent exact-rational reconstruction.
Independent review already recorded	The equalities were independently expanded; broader fold implications were excluded.

The prior closure record is evidence about the research program's internal state, not a substitute for your review. A new material objection should be reported even if the internal label is CLOSED.

2. Correction and reopening rule

Reopen or amend this object if you identify a counterexample, an object-definition mismatch, a sign/factor/quantifier defect, an unreproducible decisive computation, a hidden premise, or a scope claim not supported by the proof. Corrections must be additive and must preserve the historical source and review record.

3. Limits

- No theorem about positivity outside alpha in $[0,1]$, fold nondegeneracy, or regional counting.
- A correct lemma may still be unusable downstream if another dependency uses a different field, law, coordinate convention, branch mark, normalization, or parameter domain.
- A prior independent review does not reduce your duty to reconstruct the argument; duplicate lineages must not be counted as additional independence.

Part IV - Reproduction Instructions

- Record your name, institution or affiliation, expertise, date, and any prior exposure to this research program.
- Reconstruct the exact statement from Part I without relying on the historical verdict.
- Audit Part II line by line. For computational objects, obtain the raw carrier from the manifest, verify hashes first, inspect source safety, then execute in a clean environment.
- Complete the obligation ledger and hostile checks. Preserve exact counterexamples, code, logs, and environment details.
- Return the verdict form below plus a finding list with severity: critical, material, clarification, editorial, or no issue.

Part V - Reviewer Verdict Form

Reviewer	_____
Affiliation / expertise	_____
Prior exposure	None / limited / substantial - explain: _____
Independent reconstruction	Complete / partial / not possible
Verdict	APPROVE / APPROVE WITH CLARIFICATIONS / AMEND REQUIRED / REJECT / CANNOT VERIFY
Exact scope accepted	_____
Critical or material findings	_____ _____
Reproduction artifacts	Code/log/hash/location: _____

Signature and date _____

Required narrative: summarize your independent derivation; list every disagreement; state whether each issue affects the theorem statement, only the proof, only reviewability/reproducibility, or only presentation. State explicitly which excluded claims you did not review.

Part VI - Manifest and Provenance

Controlling source: <https://docs.google.com/document/d/1GNp80PpZ-ONaEKBllXVxDqbYyAGm35Xb-H1P2THmV6w/edit>

Closure / terminal record: https://docs.google.com/document/d/1Zm65j9BZe_z3P_8qP7rfZukRU-IWUE8h8bDqXdDsIQ/edit

Local source snapshot used to construct this packet: GP-CLS-PROP-013-v1.0 — Proposed terminal closure of the declared convex-combination identities

Local source snapshot SHA-256 (DOCX export, transport checksum only):
5ca59cbf51aca4a2039d20afcfb98f9e76b6a8117b02a0e6766d536c617e07d3

Authority note: the transport checksum above identifies the locally downloaded Office export used for packet construction. It is not substituted for any marker-delimited frozen-body hash declared inside the source artifact.

Supplemental Drive access may be granted for raw capsules, revision history, comments, exact native exports, and dependency artifacts. The packet is designed so that those materials strengthen reproducibility rather than supply missing mathematical context.